FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: 1HA-I 2311 — B(3111) / D(7114)

Retyped faithfully from the original board paper — same questions, no changes.

VERSION B

Note: MATHEMATICS HSSC-I. SECTION – A (Marks 20). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 3111. Fill the relevant bubble against each question according to curriculum:

Q1. The multiplicative inverse of –i is:

(A) i
(B) 1
(C) –i
(D) –1

Q2. What is the modulus of a complex number (8–15i)?

(A) 8+15i
(B) 17
(C) √(161)
(D) –15

Q3. The contrapositive of a conditional p→q is:

(A) q→p
(B) ~q→p
(C) ~q→~p
(D) ~p→q

Q4. Which structure in the following is true for the set of natural numbers under multiplication?

(A) Groupoid
(B) Semi group
(C) Monoid
(D) Group

Q5. Which one of the following matrices is singular?

(A) [[1,–4],[2,8]]
(B) [[1,4],[√4,8]]
(C) [[1,–4],[√2,–8]]
(D) [[–1,4],[2,8]]

Q6. Rank of matrix [[–2],[0],[–1]] is:

(A) 0 (B) 1

(C) 2 (D) 3

Q7. For what value of k, roots of $kx^2–12x+4=0$ are equal?

(A) 9 (B) –9

(C) 9.5 (D) 18

Q8. One of the multiplicative factors of $(x^4–5x^2+4)$ is:

(A) x+2 (B) x–3

(C) x+3 (D) x+4

Q9. Which one of the following represents $(x^3+2x^2+3)/((x^2+1)(x+4))$?

(A) Proper fraction (B) Improper fraction

(C) Cubic polynomial (D) Polynomial

Q10. For what value of x, the numbers 1/2, 1/5, 1/x are in harmonic progression?

(A) –10 (B) –8

(C) 8 (D) 10

Q11. If C(n,8)=C(n,12), then value of n is:

(A) 4 (B) 8

(C) 12 (D) 20

Q12. The probability of getting same upper face on throwing two fair dice simultaneously is:

(A) 1/12 (B) 1/6

(C) 1/4 (D) 1/2

Q13. The coefficient of third term in the expansion of $(x–1/x)^8$ is:

(A) C(8,0) (B) C(8,1)

(C) C(8,2) (D) C(8,3)

Q14. In which quadrant, terminal side of the angle –510° lies?

(A) 1st (B) 2nd

(C) 3rd (D) 4th

Q15. (Sin5x+Sin7x) is expressed in product form as:

(A) 2Sin6xCosx (B) 2Cos6xSinx

(C) 2Sin6xSinx (D) 2Cos6xCosx

Q16. The value of Cos(x+60°)+Cos(x–60°) is:

(A) Cosx (B) √3 Cosx

(C) Cos2x (D) 0

Q17. The period of 8/7 Sec(x–π) is:

(A) –π (B) π

(C) 3π (D) 8π/7

Q18. What is the area of a triangle ΔABC, if a=10, b=20 and γ=30°?

(A) 25√2 (B) 50√3

(C) 50 (D) 100/√3

Q19. The value of Cos[π/6 + Cos^{-1}(–1/2)] is:

(A) 1/2 (B) –1/2

(C) √3/2 (D) –√3/2

Q20. The solution of a trigonometric equation (Sinx+Cosx=0) is:

(A) –30° (B) –120°

(C) 45° (D) 135°

Note: —1HA-I 2311-3111—. Page 1 of 2. Page 2 of 2.

Note: MATHEMATICS HSSC–I. Time allowed: 2:35 Hours. Total Marks Sections B and C: 80. NOTE: Attempt any twelve parts from Section 'B' and any four questions from Section 'C' on the separately provided answer book. Write your answers neatly and legibly.

Q. SECTION – B (Marks 48)

Q. 2 Attempt any TWELVE parts. All parts carry equal marks. (12 x 4 = 48)

(i) If Z_1=2+3i and Z_2=4+2i, then show that ($Z_1\bar{Z}_2+\bar{Z}_1Z_2$) is a real number.

(ii) Construct a truth table of a logical statement (p↔q)∧(p→q).

(iii) Solve for x: det|x –1; 5 1–x| = det|1 0 –3; 2 x –6; 1 3 x–5|.

(iv) If α,β are the roots of x^2+px+q=0, find the quadratic equation whose roots are α/β and β/α.

(v) Using properties of the cube roots of unity, verify that (1+ω)+$(1+ω)^2$+$(1+ω)^3$=2ω.

(vi) Express (125+4x–$9x^2$)/((x–1)(x+3)(x+4)) in partial fractions.

(vii) Second term of a geometric sequence is 9 and its fourth term is 1. Find sum to infinity.

(viii) Insert six arithmetic means between 15 and –13.

(ix) Prove that Sinc is a periodic function and its period is 2π.

(x) A die is thrown twice. Find the probability that sum of the upper face numbers is a prime number or an odd number.

(xi) Find the value of k, if the constant term in the expansion of $(2x^2+k/x)^6$ is 960.

(xii) If $\cos\theta=\sqrt{10}/10$ with $2\pi<\theta<5\pi/2$, then find values of the remaining five trigonometric ratios.

(xiii) Verify that: $\cos4x.\cos x-\sin6x.\sin3x = \cos7x.\cos2x$.

(xiv) In an oblique triangle ΔABC (with usual notations) a=6, c=12 and β=124°. Apply law of cosines and law of sines to find the values of 'b', α and γ.

(xv) Verify that: $2\tan^{-1}(1/2)+\tan^{-1}(1/7)=\tan^{-1}(31/17)$.

(xvi) Solve the trigonometric equation $\cos5\theta+\cos\theta=\cos3\theta$ where $\theta\in[0,\pi]$.

Q. SECTION – C (Marks 32)

Note: Attempt any FOUR questions. All questions carry equal marks. (4 x 8 = 32)

Q. 3 Use Cramer's rule to solve the system of linear equations. x+y–z=3; 2x–y–z=1; 3x+y+2z=0

Q. 4 If three consecutive numbers in an arithmetic progression are increased by 1, 2 and 3 respectively, the resulting numbers are in geometric progression. Find the original numbers if their sum is 12.

Q. 5 If $y=1/4+1.3/(4.8)+1.3.5/(4.8.12)+....$, then prove that $y^2+2y-1=0$

Q. 6 Without using calculator, prove that $\cos10°.\cos30°.\cos50°.\cos70°=3/16$

Q. 7 Solve the following system of equations: $5x^2-14xy+9y^2=0$; $4x^2-3xy-16=0$

Q. 8 Solve triangles ΔABC (with usual notations) if, (a) α=60°, β=15° and b=33 (b) b=23, c=24 and α=75°

> Note: —1HA-I 2311—

VERSION D

> Note: MATHEMATICS HSSC-I. SECTION – A (Marks 20). Time allowed: 25 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7114. Fill the relevant bubble against each question according to curriculum:

Q1. What is the area of a triangle ΔABC, if a=1, c=2 and β=60°?

(A) √3 (B) √3/2

(C) 1/2 (D) 1

Q2. The value of $\sin(\pi+\sin^{-1}x)$ is:

(A) πx (B) x

(C) –x (D) –πx

Q3. Which one of the following is a solution of Sin2x+Cos2x=–1?

(A) 30° (B) 45°

(C) 90° (D) 0°

Q4. If z=–2–3i then, what is the value of z–z̄?

(A) 6i (B) –4

(C) 4 (D) –6i

Q5. The simplified form of i^{18} is:

(A) –i (B) 1

(C) –1 (D) i

Q6. The converse of the conditional ~q→~p is:

(A) ~q→p (B) q→p

(C) ~p→~q (D) q→p

Q7. Which structure in the following is TRUE for the set of natural numbers under addition?

(A) Semi group (B) Monoid

(C) Group (D) Groupoid

Q8. The determinant of a matrix [[i,0,√3],[0,i,√3],[0,0,i]] is:

(A) i (B) –1

(C) 1 (D) –i

Q9. Rank of matrix [[1],[2],[3]] is:

(A) 1 (B) 2

(C) 3 (D) 0

Q10. If (–1) is a root of $x^3+px^2-x+2=0$, then value of p is:

(A) –4 (B) 0

(C) 2 (D) –2

Q11. For what value of m, $(x^2-2mx-3)$ has zero remainder, when divided by $(x-3)$?

(A) 1 (B) –2

(C) 2 (D) –1

Q12. For what value of B, $1/(x^2-1) = A/(x-1) + B/(x+1)$?

(A) 2 (B) –1

(C) –1/2 (D) 1/2

Q13. If a–3, 6, b+3 are in arithmetic progression, then value of (a+b) is:

(A) 6 (B) 12

(C) 18 (D) 0

Q14. If ${}^{n}P_2=30$ then, value(s) of n is/are:

(A) 6,–5 (B) 8

(C) 12 (D) 6,5

Q15. The probability of getting two tails when two coins are tossed is:

(A) 1/6 (B) 1/4

(C) 1/3 (D) 1/2

Q16. The middle term in the expansion of $(x-1/(2x))^{12}$ is:

(A) 6th (B) 7th

(C) 8th (D) 5th

Q17. If $Sin^2\theta=1/7$ the value of $Sec^2\theta$ is:

(A) 6/7 (B) 7/6

(C) 8/7 (D) 7

Q18. 2Sin7xCos3x is expressed in sum or difference form as:

(A) Sin10x–Sin4x (B) Sin5x–Sin2x

(C) Cos5x+Cos2x (D) Sin10x+Sin4x

Q19. The value of Sin(90°+x)+Sin(90°–x) is:

(A) 0 (B) –2Sinx

(C) –2Cosx (D) 2Cosx

Q20. The period of 3Cos(x/5) is:

(A) 10π (B) 30π

(C) π/5 (D) 5π

> Note: —1HA-I 2311-7114 HA—. Page 1 of 2. Page 2 of 2.

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Q. SECTION – B (Marks 48)

Q. 2 Attempt any TWELVE parts. All parts carry equal marks. (12 x 4 = 48)

(i) If $Z_1=3+4i$ and $Z_2=4+3i$, then show that $|Z_1+1/Z_2| \cong 5$.

(ii) Construct truth table of a logical statement $(\sim p \rightarrow \sim q) \wedge p$.

(iii) Show that $\det|y+z \; x \; y;\; z+x \; z \; x;\; x+y \; y \; z| = (x+y+z)(z-x)^2$.

(iv) If (–5) is a root of the equation $2x^2+px-15=0$ and the equation $p(x^2+x)+k=0$ has equal roots, find values of 'p' and 'k'.

(v) Using properties of the cube roots of unity, Show that $(x+y)^2+(x\omega+y\omega^2)^2+(x\omega^2+y\omega)^2=6xy$.

(vi) Express $(x^2+2x-1)/((x)(x+2)(2x-1))$ in partial fractions.

(vii) Which term of the arithmetic sequence 4,–4,–12,–20,–28,.....is (–172)?

(viii) Show that $S_n = \Sigma(k=1 \text{ to } n)(6k^2+4k-1) = n(n+2)(2n+1)$.

(ix) Prove that Tan gent is a periodic function and its period is π.

(x) A fair cubical die is rolled three times. (a) Find the probability of getting a six all three times. (b) Find the probability of getting no sixes.

(xi) Find the value of k, if the constant term in the expansion of $(4x^2+k/(2x))^9$ is 84.

(xii) If $\text{Sin}\theta=-\sqrt{15}/15$ and $\pi<\theta<3\pi/2$, then find values of the remaining trigonometric ratios.

(xiii) If $\text{Cosec}\alpha=5/3$, where $0<\alpha<\pi/2$ and $\text{Sec}\beta=13/12$, where $3\pi/2<\beta<2\pi$, then find the value of $\text{Cos}(\alpha-\beta)$.

(xiv) Two escribed circles of radii r_2 and r_3 are connected with triangle ΔABC. Using usual notations, prove that $(r_2+r_3)\tan(\alpha/2)=a$.

(xv) Prove that: $2\tan^{-1}(1/3)+\tan^{-1}(1/7)=\pi/4$.

(xvi) Solve the trigonometric equation $\text{Sin}2x=\sqrt{3}\text{Cos}x$, where $0<x<2\pi$.

Q. SECTION – C (Marks 32)

Note: Attempt any FOUR questions. All questions carry equal marks. (4 x 8 = 32)

Q. 3 Use Cramer's rule to solve the system of linear equations x+5y–3z=–36; x+4y+2z=–11; 2x–y=7

Q. 4 If 1, 4 and 3 are added to three consecutive terms of a geometric progression, the resulting numbers are in arithmetic progression. What are the numbers if their sum is 13?

Q. 5 If $y=3/4+3.5/(4.8)+3.5.7/(4.8.12)+....$, then prove that $y^2+2y-7=0$

Q. 6 Without using calculator, prove that $Cos40°.Cos80°.Cos120°.Cos160°=1/16$

Q. 7 Solve the following system of equations $6x^2+9xy+3y^2=0$; $x^2-5xy+6y^2=26$

Q. 8 Solve triangle ΔABC (with usual notations) if, (a) $a=7$, $b=10$ and $c=13$ (b) $a=25$, $b=24$ and $\gamma=120°$

> Note: —1HA-I 2311 HA—